Jedha

CDSD block 1

Cyril Braguy



Project KAYAK

Github project : cyrilbraguy/Bl1_kayak

Kayak application on Huggingface : Kayak_trip_application

Goal of the project

Pitch

Design a travel application, a search engine
to recommend the best trip destinations within a list of 35 cities to visit in France,
from reliable informations on :

Best weather forecast,

hotel rooms availability in these destinations

Goal

- Get info on cities : GPS coordinates
- Get weather data of cities at 5 days
  - order cities bv weather results
  - select best 5 cities
- Get hotels info about each destination
- Store all the information above in a data lake : one file .csv
- Extract, transform and load cleaned data from your datalake to a data warehouse
- Display these informations on a nice frontend

Detailed pipeline :

1. Cities + weather

- 35 cities list → get lat & lon from nominatim.org → dataFrame
- Get weather data of cities at 5 days in 3 hrs slots from openweathermap.org [D to D+4]
- Average weather data on each time window eg : [D+1 – D+4] & [D+3 – D+4]
  - groupby data on each day (average, sum, max)
  - Compute a **weather comfort score**
  - order cities by **weather comfort score**
  - select best 5 cities : weather_avg[‘selected’] =1
- Store in a .csv file on S3

Detailed pipeline :
2. Hotels

- booking_scraper.py : A special tool developed to scrape booking.com : using playwright
  - [city, checkin, checkout, nb_adults, nb_children, nb_rooms] → 5 hotels w/ several informations :
  - [hotel_name,url_hotel,address_hotel,hotel_description,lat_hotel,lon_hotel,score_hotel,price_hotel,currency_hotel,room_description,hotel_city,combined_score_hotel]
- Combined_score = score_hotel - p * (price_hotel - max_price).clip(lower=0)
- Store all the information above in a data lake : **hotels_top20_cities.csv**

Detailed pipeline

3. ETL & SQL database

- Load last **hotels_top20_cities.csv**
- Save information in 3 sql tables to respect unicity principle :
  cities weather hotels

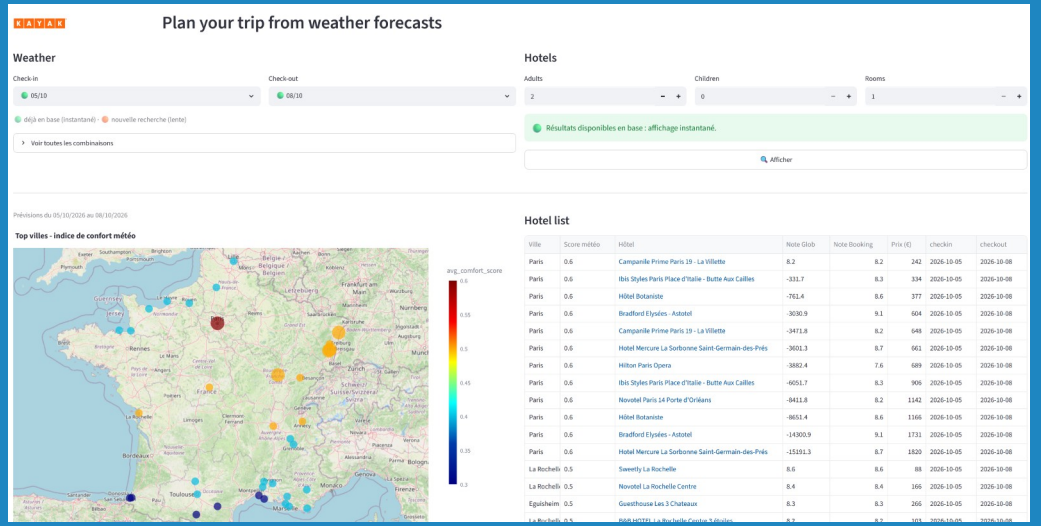
table	key_id	columns
cities	city_id	city, city_id, city_lat, city_lon, city_name_, city_plus
weather	city_id ; checkin_date ; checkout_date	city_id, city, checkin_date, checkout_date, avg_comfort_score, ... selected, lat, lon
hotels	city_id ; checkin_date ; checkout_date	city_id, city, checkin_date, ... combined_score_hotel

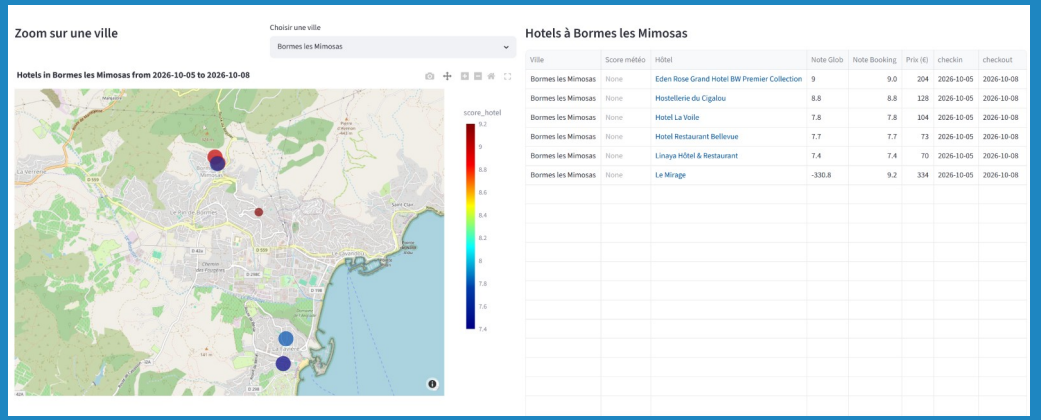
Detailed pipeline

4. Frontend

A streamlit frontend is used

- Reads sql database and displays filtered and ordered data
- 1 : weather & best cities
  displays best cities w/ color map & bigger marker size
  & hotels list by order of
- 2 : select one cities ; view hotels list order by combined_score





Complete solution

**An application in a Docker container
in huggingface space**

- 1 : pipeline linear get weather hotels ;
  ETL : cron 6h & 18 h ; store in S3 & neon SQL dB
- 2 : Streamlit frontend to print results for customer
- Git repo for code store & version
- Github_action to deploy in HuggingFace space at push
- S3 bucket Sql database on NeondB